AS · Unit 2 · WBI12 Cells, Development, Biodiversity and Conservation

Cell structure

Every organism is built from cells, and inside each cell different structures carry out different jobs. This page covers the ultrastructure of eukaryotic and prokaryotic cells, the route a secreted protein takes through the cell, how cells are grouped into tissues and organs, and how microscopes, stains and simple calculations let us measure what we see.

SPEC 3.1 All living organisms are made of cells

WHAT THIS SPEC POINT ASKS

Know that every living organism is made of one or more cells, and that all cells share some basic features.

Cell theory

- Every living organism is made of **one or more cells**. Unicellular organisms (bacteria, yeast, *Amoeba*) are a single cell; multicellular organisms are made of millions.
- The cell is the **smallest unit** that can carry out all the processes of life (respiration, growth, reproduction, response…).
- New cells only come from **existing cells**, by cell division.

Features every cell has

Feature	Why every cell needs it
Cell surface membrane	separates the cell from its surroundings and controls what enters and leaves
Cytoplasm	where most metabolic reactions happen
DNA	the genetic material: codes for the cell's proteins and is copied when the cell divides
Ribosomes	the site of protein synthesis (translation)

Two kinds of cell

	Prokaryotic (bacteria and archaea)	**Eukaryotic** (animals, plants, fungi, protoctists)
Nucleus	**none**: DNA lies free in the cytoplasm	DNA inside a **nucleus** with a double membrane
DNA	**circular**, not wound round histone proteins; often **plasmids** too	**linear** chromosomes wound round **histones**
Membrane-bound organelles	**none**	yes: mitochondria, ER, Golgi, lysosomes (and chloroplasts in plants)
Ribosomes	smaller, **70S**	larger, **80S** in the cytoplasm
Typical size	about 1–5 μm	about 10–100 μm
Cell wall	always: **peptidoglycan (murein)**	plants: cellulose; fungi: chitin; animals: none

- **Ultrastructure** means the detail of a cell that you can only see with an **electron microscope** (3.3, 3.6).
- The two main groups also show in classification: prokaryotes make up the domains **Bacteria** and **Archaea**; every eukaryote is in the domain **Eukarya** (more in 4.14).

MARK-SCHEME WORDING

- Eukaryote clue: "this organism has a **nucleus**" / "prokaryotic cells do not contain a nucleus". Cell membrane and cytoplasm don't count: every cell has them.
- Domain: "**Eukarya**" (eukaryote / eukaryota accepted). "Prokarya" and "Plantae" are not domains.
- Tick-grids: cell surface membrane, cytoplasm and ribosomes are in **animal, plant and prokaryotic** cells. Cell wall: **plant and prokaryotic** only. Nucleolus: **not** in prokaryotes.

Often asked as: tick-grids of structures in animal, plant and prokaryotic cells; name the domain of an organism; give one reason an organism is not a prokaryote.

COMMON MISTAKE

"Bacteria have a nucleus" or "bacterial DNA is linear". Bacterial DNA is circular and lies free in the cytoplasm.

Naming "plants" or "Archaea" as the domain of a single-celled organism with chloroplasts. A chloroplast is a membrane-bound organelle, so the organism is in the **Eukarya**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Circular DNA in eukaryotic cells: mitochondria and chloroplasts carry their own small **circular DNA** and **70S ribosomes**. So "circular DNA" and "ribosomes" are ticked for animal, plant **and** prokaryotic cells (WBI12 Jan 2024). Few candidates knew this.

SPEC 3.2 Cells, tissues, organs and organ systems

WHAT THIS SPEC POINT ASKS

Understand how the cells of a multicellular organism are grouped into tissues, tissues into organs, and organs into organ systems.

The levels

1. **Specialised cell**: a cell whose structure suits one job (e.g. a red blood cell carrying O_2). Cells become specialised by **differential gene expression** (3.18).
2. **Tissue**: a group of **similar cells** (the same cell type) working together to carry out a particular function, sometimes with material between the cells.
3. **Organ**: a group of **different tissues** working together to carry out one or more functions.
4. **Organ system**: a group of **organs** working together to carry out a major function.

Examples at each level

Level	Animal example	Plant example
Specialised cell	red blood cell, sperm, ciliated epithelial cell	palisade mesophyll cell, root hair cell, guard cell
Tissue	squamous epithelium, cardiac muscle, blood (a connective tissue), nervous tissue	**xylem, phloem**, palisade mesophyll, spongy mesophyll, sclerenchyma, meristem
Organ	heart (cardiac muscle + connective tissue + epithelium + nervous tissue), small intestine, lung	leaf, stem, root, flower
Organ system	circulatory, digestive, gas exchange, nervous	shoot system, root system

The four main animal tissue types are **epithelial** (lines surfaces), **connective** (blood, bone, cartilage), **muscle** and **nervous**.

MARK-SCHEME WORDING

- Tissue: "group of (similar) cells working together to perform a (specific) function". Cells with a similar structure, function or origin are accepted.
- Organ: "(group of) tissues which work together to perform {a function / functions}". "Same tissues" is not accepted.
- Organ system: "group of organs that work together to perform {one / one or more / specific} functions".
- Organ v tissue: "an organ has many functions **whereas** a tissue has one (or fewer)"; "an organ has several tissues / cell types **whereas** a tissue has one cell type / similar cells". Two separate definitions are not pieced together.
- Plant tissue: xylem, phloem, meristem, sclerenchyma, parenchyma, palisade or spongy mesophyll. "Palisade mesophyll **cell**" is rejected: that's a cell.

Often asked as: one-mark definitions of tissue, organ or organ system; two differences between an organ and a tissue; name a plant tissue.

COMMON MISTAKE

"A tissue is a group of **specialised** cells." Too vague: the mark needs **similar cells / the same type of cell**.

Writing "tissues" for an organ system, or giving the tissue definition for an organ.

Naming a cell, a molecule or an organ when a tissue is asked for.

SPEC 3.3 Eukaryotic cell ultrastructure and organelle functions

WHAT THIS SPEC POINT ASKS

(i) Know the ultrastructure of eukaryotic cells: nucleus, nucleolus, ribosomes, rough and smooth endoplasmic reticulum, mitochondria, centrioles, lysosomes and Golgi apparatus. (ii) Understand what each of these does.

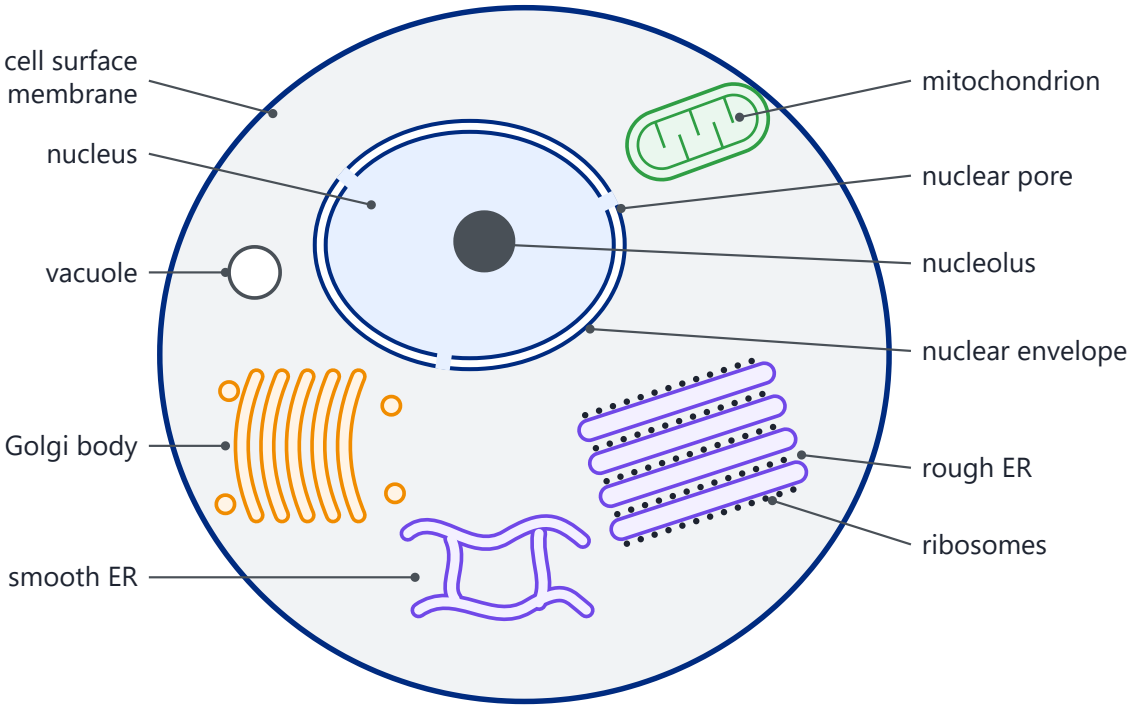
Organelle → structure → function

Organelle	Structure	Function
Nucleus	largest organelle (about 10–20 μm). **Nuclear envelope** = double membrane with **nuclear pores**. Contains **chromatin** (DNA wound round histones) in the nucleoplasm	holds the genetic material; site of **DNA replication** and **transcription**. Pores let mRNA and ribosome subunits out, and nucleotides and proteins in

Organelle	Structure	Function
Nucleolus	dense, dark region inside the nucleus; **no membrane**; made of rRNA, protein and the DNA that codes for rRNA	makes **rRNA** and assembles **ribosome subunits**
Ribosomes	tiny (about 25 nm), **no membrane**; a large and a small subunit made of rRNA and protein; **80S** in the cytoplasm. Free in the cytoplasm or attached to rER	site of **protein synthesis** (translation). Free ribosomes make proteins used inside the cell; rER ribosomes make proteins for **secretion**, membranes or lysosomes
Rough ER (rER)	flattened membrane sacs (**cisternae**) continuous with the nuclear envelope; outer surface **studded with ribosomes**	proteins made on its ribosomes enter the space inside, **fold** into their 3D shape and are **transported** in vesicles to the Golgi (3.4)
Smooth ER (sER)	membrane tubes and sacs with **no ribosomes**	makes and processes **lipids** (phospholipids, cholesterol) and **steroid hormones**
Golgi apparatus	a stack of flattened, **curved** membrane sacs (cisternae), **not joined** to each other, with **vesicles** budding off the edges; no ribosomes	**modifies** proteins (e.g. adds carbohydrate to make glycoproteins), then **packages** them into vesicles: secretory vesicles for **exocytosis**, or **lysosomes**
Mitochondrion	**double membrane**; inner membrane folded into **cristae** (large surface area); fluid **matrix** with enzymes, its own circular DNA and 70S ribosomes; about 1–10 μm long	site of **aerobic respiration**, which produces **ATP**. Many in very active cells (muscle, sperm midpiece, secretory cells)
Lysosome	small vesicle with a **single membrane**, containing **hydrolytic (digestive) enzymes**; no internal structure; made by the Golgi	**digests** worn-out organelles, pathogens taken in by phagocytes, or whole dead cells. The membrane keeps the enzymes away from the rest of the cell
Centrioles	a pair of short hollow **cylinders** at right angles, each made of **nine triplets of microtubules**; near the nucleus; in animal cells (not most plant cells)	organise microtubules to form the **spindle fibres** in mitosis and meiosis, which **separate chromosomes/chromatids**

The **cell surface membrane** (a phospholipid bilayer with proteins, from Unit 1) surrounds all of this.

Animal v plant cells: plant cells also have a **cellulose cell wall**, **chloroplasts**, a **large permanent vacuole** (with its tonoplast), amyloplasts and plasmodesmata, and usually no centrioles. That is Topic 4 (4.1).



An animal cell as it appears in an electron micrograph. Note the double nuclear membrane with pores, ribosome-studded rough ER, and the stacked sacs of the Golgi body.

EXAM TIP

"Function" questions want a **verb plus a product**: "aerobic respiration to make ATP", "makes ribosomes", "forms spindle fibres", "contains hydrolytic enzymes that digest pathogens". A description of the structure scores nothing when function is asked.

Drawing a nucleus: **two lines** for the envelope with **gaps** for pores, a dark **nucleolus**, patches of **chromatin**. Draw only the nucleus, not the whole cell.

MARK-SCHEME WORDING

- Mitochondrion: "**aerobic respiration**", "produces **ATP**" (Krebs cycle, oxidative phosphorylation accepted). "Produces energy" is **not** accepted.
- Nucleolus: "produces **ribosomal subunits**" (ribosomes / rRNA accepted). "Contains genetic material" is ignored.

- Nucleus: "contains the genetic material / DNA", "DNA replication", "transcription". Plain "controls the cell" is ignored.
- Centriole: "forms **spindle fibres** in mitosis / meiosis" (forms microtubules accepted) "to separate chromosomes / chromatids". Centromere for centriole is rejected.
- Lysosome: "contains **digestive / hydrolytic** enzymes" or "digests pathogens / organelles". Plain "enzymes" is ignored.
- Drawing the rER: at least two tubules with **ribosomes on the surface**; cisternae and ribosomes labelled. Drawing a nucleus: double membrane, pores as gaps, nucleolus, chromatin. One wrong label caps the label marks.
- Two structures inside the nucleus: "nucleolus" and a named genetic material (DNA, chromatin, chromosomes). The envelope, pores and nucleoplasm aren't credited.
- Prophase events: chromosomes condense, nuclear envelope breaks down, centrioles move to the poles and produce spindle fibres.

Often asked as: name an organelle from a description or EM and give its function; draw and label a nucleus, rER or Golgi; compare a lysosome with an acrosome; suggest why mitochondria sit next to lipid droplets.

COMMON MISTAKE

"Mitochondria produce energy." Energy can't be produced: say "**aerobic respiration**, which produces/releases **ATP**".

Confusing the **nucleolus** with the nucleus, the **centriole** with the centromere, or rER with sER (look for ribosomes). Saying the Golgi "makes proteins": it **modifies and packages** them.

Label lines that stop in the cytoplasm instead of touching the structure.

In a compare question, writing a paragraph about each organelle. Use paired sentences: "both have…; a lysosome… **whereas** an acrosome…".

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Chloroplasts and mitochondria **both** have a double membrane and both contain ribosomes and circular DNA (WBI12 Jan 2026).
- **Lysosome v acrosome**: both are single-membrane vesicles of hydrolytic enzymes made by the Golgi; a lysosome digests **inside** the cell, the acrosome digests the zona pellucida **outside** it (WBI12 Jun 2023).
- Mitochondria next to **lipid droplets** in an egg: a **short diffusion distance**, so fatty acids reach the mitochondria quickly for respiration to make ATP (WBI12 Jan 2021).
- The number of organelles roughly **doubles in interphase** before a cell divides, so each daughter cell gets a full set (WBI12 Jun 2021).

SPEC 3.4 The rER and Golgi in protein transport

WHAT THIS SPEC POINT ASKS

Understand how the rough ER and the Golgi apparatus move proteins through a cell, including how extracellular enzymes are made and released.

The secretory pathway, step by step

Follow one protein, e.g. **amylase** made by a pancreas cell:

1. **Nucleus**: the gene is **transcribed** into mRNA, which leaves through a **nuclear pore**.
2. **Ribosomes on the rER**: the mRNA is **translated**; the growing polypeptide passes into the **rER lumen**.
3. **rER**: the polypeptide **folds** into its secondary and tertiary structure (hydrogen, ionic and disulfide bonds form). It is **packaged into a transport vesicle** that buds off the rER.
4. The vesicle moves to the **Golgi** and **fuses** with it, so the protein enters the Golgi.
5. **Golgi**: the protein is **modified** (for example, carbohydrate is added to make a **glycoprotein**), then **sorted and packaged** into **secretory vesicles** that bud off.
6. Secretory vesicles move to the **cell surface membrane**, **fuse** with it and release the enzyme outside the cell by **exocytosis**.

Mitochondria supply the **ATP** for moving vesicles and for exocytosis.

Extracellular enzymes

- An **extracellular enzyme** works **outside the cell** that made it: digestive enzymes (amylase, trypsin, lipase), enzymes that fungi release to digest wood, the enzymes a Venus flytrap or pitcher plant secretes onto its prey.
- Cells that secrete a lot of protein (pancreas, salivary glands, plasma cells) are packed with **rER, Golgi** and mitochondria.
- The Golgi sends proteins to other places too: **lysosomes** (enzymes kept inside the cell), and proteins built into the **cell surface membrane**.

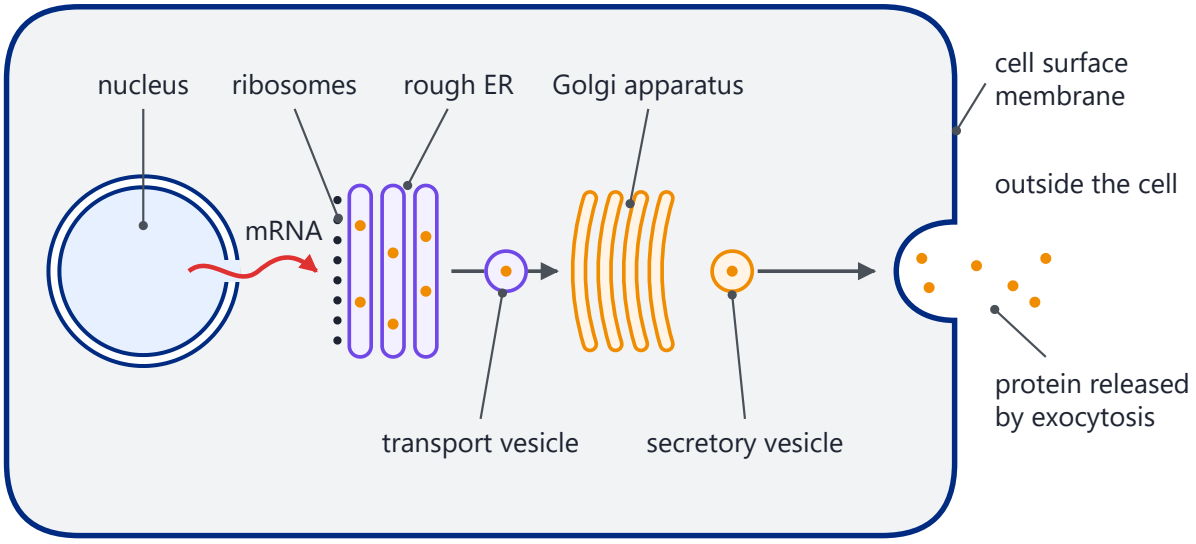
WORKED EXAMPLE

Evidence: following labelled amino acids

Pancreas cells are given radioactive (or fluorescent) amino acids for a short time, then samples are taken over time.

The label appears first in the **rER**, then the **Golgi**, then **secretory vesicles**, and finally **outside the cell**.

Explain it: the amino acids are built into polypeptides on the rER ribosomes; the protein moves in vesicles to the Golgi, is modified and packaged, and leaves by exocytosis. Inside the cell the label falls as outside it rises.



The secretory pathway, from transcription in the nucleus to release of the finished protein at the cell surface.

The route of a secreted protein

1. In the nucleus, the gene for the protein is transcribed into mRNA, which leaves through a nuclear pore.
2. The mRNA attaches to a ribosome on the rough ER. As the polypeptide is made, it is fed into the lumen of the RER.

   Making it straight into the RER keeps the protein separate from the cytoplasm, so it can be packaged for export.
3. Inside the RER the polypeptide folds into its secondary and tertiary structure, and it moves through the ER sacs.
4. Small transport vesicles pinch off the RER, carrying the protein. They move to the Golgi apparatus and fuse with its cis face.
5. In the Golgi apparatus the protein is modified, for example by adding carbohydrate to form a glycoprotein, and then sorted.
6. Secretory vesicles bud from the trans face of the Golgi and move along the cytoskeleton to the cell surface membrane.

7 The vesicle membrane fuses with the cell surface membrane and the protein is released outside the cell. This is exocytosis.

Exocytosis needs ATP, which is why secretory cells contain many mitochondria.

EXAM TIP

Use the protein's **name** from the question ("insulin", "the enzyme", "acrosin") at every step. Marks for the link are lost if you write just "protein". If the question gives you the polypeptide, **don't start with transcription**: start where the question starts.

MARK-SCHEME WORDING

- Route (5 marks, insulin): "rER packages insulin in a vesicle"; "vesicle **fuses** with the Golgi apparatus" and insulin is modified; "Golgi packages insulin in (secretory) vesicles"; "vesicles fuse with the cell surface membrane"; "exits by **exocytosis**".
- rER alone: "translation at rER ribosomes / polypeptide enters the rER"; "forms secondary / tertiary / 3D structure" with named bonds; "packaged into vesicles to go to the Golgi".
- Golgi alone: "**modification** of the protein (into the enzyme)"; "packaged into secretory vesicles"; "leaves the cell by exocytosis".
- Glycoprotein: "**carbohydrate added** (in the Golgi)" (glycosylation accepted). "Modified in the Golgi" with no carbohydrate is ignored.
- Drawing a Golgi: at least two **curved cisternae**, no ribosomes or connections, a vesicle, cisternae labelled.
- Tracer data: label "decreases inside the cell / increases outside"; "vesicles fuse with the cell surface membrane and release…". Amino acids themselves leaving is ignored.

Often asked as: describe how an enzyme or hormone is made and secreted (3–5 marks, often in a plant, fungus or sperm context); the role of the rER or Golgi alone; explain tracer data; draw a Golgi.

COMMON MISTAKE

Giving the Golgi's role when the rER was asked, or the reverse. Saying **amino acids** are exocytosed: the finished protein is.

Writing "moves to the Golgi". The mark is for the vesicle **fusing** with the Golgi, and this is the step most often missed.

Writing "proteins" when the question is about **enzymes**; mixing up transcription and translation; writing lipids for proteins.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Glycoproteins** are made in the Golgi by adding carbohydrate chains to a protein, e.g. the Rhesus antigen (WBI12 Oct 2022) and mucus.
- The rER vesicle **fuses** with the Golgi (at its "cis" face, where vesicles arrive); secretory vesicles bud from the opposite "trans" face (WBI12 Jan 2020).
- Enzymes in the egg's **cortical granules** are made by the same rER → Golgi route (WBI12 Oct 2024).
- Role of one organelle in making acrosin: nucleus (transcription), ribosome (translation), rER (folding, packaging), Golgi (modification, packaging) or mitochondria (ATP). The **acrosome** itself is not credited: it only stores the enzyme (WBI12 Jun 2022).

SPEC 3.5 Prokaryotic cells

WHAT THIS SPEC POINT ASKS

(i) Know the ultrastructure of prokaryotic cells: cell wall, capsule, plasmid, flagellum, pili, ribosomes and circular DNA. (ii) Understand what each structure does.

Structure → function

Structure	What it is	Function
Cell wall	rigid layer of **peptidoglycan (murein)**: polysaccharide chains cross-linked by short peptides	stops the cell **bursting** when water enters by osmosis; keeps its shape
Capsule (slime layer)	layer of polysaccharide **outside** the cell wall	protects against **drying out**, against being engulfed by **phagocytes**, and against antibodies; helps bacteria **stick** to surfaces and to each other
Cell surface membrane	phospholipid bilayer	controls what enters and leaves; some respiration enzymes are on it
Circular DNA	one long **circular** DNA molecule, **not bound to histones**, free in the cytoplasm (the nucleoid region)	carries the genes for almost all the cell's proteins
Plasmids	small extra rings of DNA that copy themselves independently	carry **extra genes**, e.g. for **antibiotic resistance**; can be passed between bacteria

Structure	What it is	Function
Ribosomes	**70S**: smaller than eukaryotic 80S ribosomes	protein synthesis (translation)
Flagellum	long protein filament attached at the cell membrane, **rotated** by a motor at its base	**movement** (motility) through liquid
Pili	short, thin protein threads on the surface	**attachment** to surfaces and host cells; a **sex pilus** joins two bacteria for **conjugation** (plasmid transfer)

- **Always present**: cell wall, cell surface membrane, cytoplasm, circular DNA, 70S ribosomes.
- **Only some bacteria**: capsule, plasmids, flagella, pili.
- **Never present**: nucleus, nucleolus, mitochondria, ER, Golgi, chloroplasts or any other membrane-bound organelle.

MARK-SCHEME WORDING

- Capsule: "prevents **dehydration** of the cell"; protection from **phagocytes / antibodies / antibiotics** accepted; helps adhesion. Plain "protection" is too vague.
- Pili: "allow bacteria to **adhere** to surfaces"; conjugation / exchange of plasmids accepted. "Sexual reproduction" is ignored.
- Flagellum: "for (cell) **motility** / movement".
- Plasmid: "contains **genes** to make proteins", e.g. antibiotic resistance. Say what it **does**, not just what it is.
- Ribosomes: "**70S** whereas eukaryotic ribosomes are **80S**", or "smaller than eukaryotic ones". You need **both halves** of the comparison.
- Structures that make an enzyme in a bacterium: "chromosome / circular DNA **and** (70S) ribosomes". Both are needed. The nucleus, rER or Golgi are wrong.
- Wall molecule: plant "cellulose"; prokaryote "**peptidoglycan / murein**".
- Bacteria using O_2: "for aerobic respiration / to produce ATP". "Produce energy" is rejected.

Often asked as: state the function of a capsule, pilus, flagellum or plasmid; draw and label two of them on an outline; how prokaryotic ribosomes differ; compare a fungal cell with a bacterium.

COMMON MISTAKE

Drawing the capsule or pili **inside** the cell wall, or a flagellum that doesn't start at the membrane.

Confusing a **plasmid** with **plasmodesmata** (channels between plant cells).

Writing **chitin** for the bacterial wall (chitin is fungal). In a compare question, wrong statements cancel right ones, so leave out anything you're unsure of.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **70S v 80S**: "S" is a measure of how fast the particle sinks in a centrifuge, so bigger means larger. Prokaryotes have 70S ribosomes; eukaryotes have 80S in the cytoplasm but 70S inside mitochondria and chloroplasts (WBI12 Jan 2020, Oct 2022).
- **Fungus v bacterium**: a fungal cell has a nucleus, linear DNA, membrane-bound organelles, 80S ribosomes and a **chitin** wall (WBI12 Jan 2025).
- A bacterium living on dry skin doesn't need a flagellum: it isn't in a liquid, so it can't swim anyway (WBI12 Oct 2024).

SPEC 3.6 Recognising organelles in electron micrographs

WHAT THIS SPEC POINT ASKS

Be able to identify the organelles listed in 3.3 in electron microscope (EM) images.

What to look for

Organelle	How it looks in a TEM	Don't confuse it with
Nucleus	the largest structure; a **double** envelope (two close lines, broken by **pores**); grainy, dark patches of chromatin	a vacuole (single membrane, no contents)
Nucleolus	the darkest, densest round area **inside** the nucleus; no membrane round it	the nucleus itself
Mitochondrion	oval, sausage-shaped or round profiles about 1 μm across; **double** membrane; inner membrane folded into shelves (**cristae**); grainy matrix	a chloroplast (bigger, with stacks of membranes called grana)
rER	parallel flattened sacs whose outer surfaces are **dotted with ribosomes**; often near the nucleus	the Golgi (no dots)
sER	tubes and sacs with **no dots**	rER
Golgi apparatus	a stack of **curved**, flattened sacs, not joined, with small round **vesicles** at the edges; no ribosomes	rER
Ribosomes	tiny dark dots, free in the cytoplasm or on the rER	—

Organelle	How it looks in a TEM	Don't confuse it with
Lysosome	small, round, **dark** body with a **single** membrane and no internal structure	a secretory vesicle (usually paler)
Centrioles	two short cylinders at right angles: one cut lengthwise looks like a striped rectangle; one cut across looks like a **ring of nine** tiny groups	—

Why the same organelle can look different

- An electron micrograph shows a **thin slice**. A 3D organelle cut **across** looks round; cut **lengthwise** it looks long. Two photos of the same organelle can differ because of the **angle (plane) of the cut**.
- EM images are black and white. Colour in a published EM is **false colour**, added afterwards.
- Fine internal detail (cristae, ribosomes, membranes) means the image came from an **electron microscope**. A 3D surface view means a **scanning** EM (3.7).

MARK-SCHEME WORDING

- Different appearance: "viewed from a different angle / one is a transverse and one a longitudinal section"; "depends on where the cell was cut".
- Which microscope: "**electron** microscope" because of the "**high resolution**", so (ultra)structure can be seen. A 3D image of a pollen grain's surface: **scanning** EM (TEM rejected). "Image is clear" is ignored.
- Chloroplast function: "site of photosynthesis / light absorption". "Produces ATP" alone and "stores starch" are ignored.

Often asked as: name or label structures on an EM; suggest why two images of one organelle differ; state which type of microscope was used and why.

COMMON MISTAKE

Saying two images of the same organelle differ in **magnification, resolution or staining** when they come from the same photo: it's the **plane of the section**.

Choosing a light microscope because the image is 2D or black and white.

Failing to recognise centrioles (often called mitochondria or lysosomes) or a chloroplast.

SPEC 3.7 Magnification, resolution and staining

WHAT THIS SPEC POINT ASKS

(i) Know how light and electron microscopes achieve magnification and resolution. (ii) Understand why specimens are stained.

Two different ideas

- **Magnification**: how many times bigger the image is than the real object.
- **Resolution**: the ability to **distinguish between two points that are close together** as separate points. The smaller the distance, the higher the resolution and the more detail you see.
- Magnifying beyond the resolution just makes a **bigger blur**. Detail comes from resolution, not magnification.

magnification = image size ÷ actual size

Rearranged: **actual size = image size ÷ magnification**; image size = actual size × magnification. Both sizes must be in the **same units**.

Convert	Multiply by
mm → µm	**1000**
µm → nm	1000
cm → µm	**10 000**

Light v electron microscopes

	Light microscope	Transmission EM (TEM)	Scanning EM (SEM)
Uses	light, focused by glass lenses	a beam of **electrons** passing **through** the specimen, focused by electromagnets	electrons **scanned across the surface**; scattered electrons form the image
Resolution	about **200 nm**	about **0.5 nm** (best)	about 3–10 nm
Useful magnification	up to about ×1500	over ×500 000	lower than TEM (about ×100 000)
Image	colour, 2D	black and white, 2D, **internal** detail	black and white, **3D surface**

	Light microscope	Transmission EM (TEM)	Scanning EM (SEM)
Specimen	living or dead	dead, very **thin** section, in a **vacuum**	dead, in a vacuum; needn't be thin

- **Why electrons resolve more**: resolution is limited by **wavelength**. Light has a wavelength of about 400–700 nm, so points closer than about 200 nm blur together. Electrons have a **much shorter wavelength**.
- Drawbacks of EM: specimens must be dead (vacuum), preparation is complex, and preparation can create **artefacts** (structures that weren't there in the living cell).

Why stain?

- Most cell structures are **colourless and transparent**, so there is little **contrast** between them.
- A stain binds to particular molecules or structures and colours them, so they stand out from their surroundings. Using two stains that bind different things (**differential staining**) separates structures.
- Staining improves contrast, **not** resolution or magnification.

Stain	What it shows
Methylene blue	nuclei (DNA) blue, e.g. cheek cells (CP5)
Acetic orcein / toluidine blue	**chromosomes**, so stages of mitosis can be identified (CP6)
Iodine solution	starch blue-black
Phloroglucinol + acid, or safranin	lignin red (xylem, sclerenchyma; safranin also stains nucleic acids)
Nile red	lipids
Eosin	cytoplasm pink

For electron microscopes, specimens are "stained" with **heavy metal** salts, which absorb electrons and show up dark.

WORKED EXAMPLE

magnification from a scale bar

A 2 µm scale bar on a micrograph measures **18 mm** with a ruler.

Convert to the same unit: 18 mm × 1000 = 18 000 µm.

Magnification = 18 000 ÷ 2 = **×9000** (write no unit).

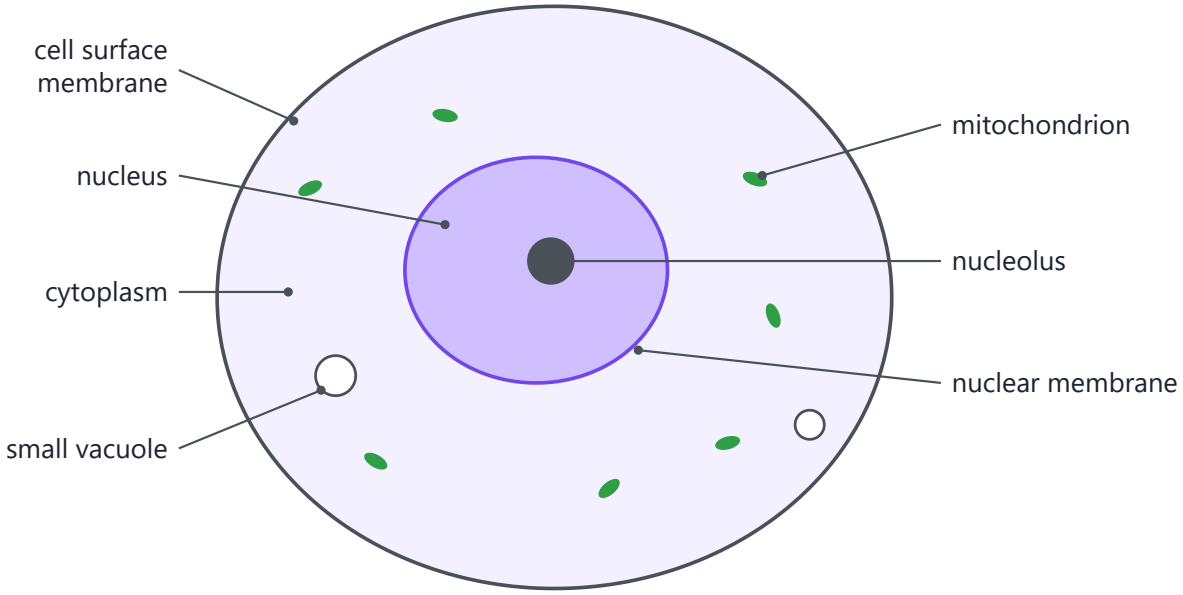
A mitochondrion in the same image is 27 mm long. Actual length = 27 000 µm ÷ 9000 = **3 µm** (= 3 × 10^{-3} mm = 3000 nm).

WORKED EXAMPLE

actual size

A cell in a drawing is **45 mm** across; the magnification is ×1500.

Actual size = 45 000 µm ÷ 1500 = **30 µm**.



The same kind of cell seen with a light microscope after staining. The nucleus shows clearly, but mitochondria are only small dots and the ER, Golgi and ribosomes cannot be made out.

Worked example: actual size from a micrograph

1. A mitochondrion measures 36 mm long on an electron micrograph with a magnification of ×12 000. Find its real length in µm.

2. Convert the image length to µm: 36 mm × 1000 = 36 000 µm.

   Converting first avoids answers that are out by a factor of 1000.

3. Actual size = image size ÷ magnification = 36 000 ÷ 12 000 = 3 µm.

4. Check: 3 µm is a sensible length for a mitochondrion (usually 1 to 10 µm).

Worked example: magnification from a scale bar

1. A scale bar labelled 5 µm is 25 mm long on the printed image.
2. Convert the bar length: 25 mm = 25 000 µm.
3. Magnification = 25 000 ÷ 5 = ×5000.

A scale bar is just an object of known size, so the same equation applies.

MARK-SCHEME WORDING

- Magnification: measure the **scale bar** (not the specimen), convert mm to µm (×1000), divide by the scale bar's value. **No unit**. Usually ±1 mm tolerance; follow the s.f. or d.p. asked.
- Resolution: "the ability to distinguish between two points which are close together".
- Magnification: "how many times bigger an object appears in an image than in real life". The equation alone is ignored.
- Why a structure can't be seen with a light microscope: "the **resolution** (or magnification) of a light microscope is not high enough" **and** "the structure is very small". For the membrane bilayer: "the two layers are very close together".
- Why an EM image shows more: "more structures can be seen as **separate** due to the **higher resolution**". Plain "high resolution" is ignored.
- Why stain: "to increase **contrast**", "to make the **chromosomes / chromatids** visible" (to identify stages of mitosis). "To make cells / organelles visible" is ignored.

Often asked as: calculate a magnification from a scale bar (almost every paper); calculate an actual size; explain why a structure can't be seen with a light microscope; define resolution; why stain.

COMMON MISTAKE

Converting cm to µm by ×1000 (it's ×10 000). Inverting the formula. Putting µm after a magnification.

Giving 2 d.p. when 2 s.f. was asked.

Saying an EM shows more because it has **higher magnification**. The reason is **resolution** (shorter wavelength).

Stating the equation without rearranging it when actual size is asked.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Named stains**: acetic orcein (WBI12 Jan 2020); safranin (lignin, nucleic acids) v nile red (lipids) (WBI12 Oct 2025).
- **Sphere volume** $V = \frac{4}{3}\pi r^3$ (formula given): use the **radius**, half the diameter (WBI12 Jun 2025). **SA:V** of a cube: $SA = 6l^2$, $V = l^3$, give it as **x : 1** (WBI12 Oct 2021).
- Unit conversions inside a rate: 15 mm at 0.2 µm s^{-1} = 15 000 ÷ 0.2 = 75 000 s (WBI12 Jan 2024).

SPEC 3.8 Core Practical 5: light microscopy, drawings and graticules

WHAT THIS SPEC POINT ASKS

(i) Use a light microscope to observe suitable animal cells and make labelled drawings. (ii) Use an eyepiece graticule to measure cells, and understand scale.

The two scales

- The **eyepiece graticule** is a scale of 100 equal divisions inside the eyepiece. It has **no units**: its divisions (eyepiece units, epu) cover a different real length at each objective.
- The **stage micrometer** is a slide with a scale of known length (often 1 mm divided into 100 divisions, so each is **10 µm**; check the slide).
- Calibrating means finding out how many µm **one eyepiece division** is worth at the objective you're using.

CORE PRACTICAL

Core practical 5: method in brief

Calibrate

1. Put the stage micrometer on the stage and focus on it with the low-power objective.
2. Turn the eyepiece until its graticule lies alongside the micrometer scale, with the zeros lined up.
3. Find a point further along where the two scales line up again. Note the number of eyepiece divisions and the length on the micrometer.
4. 1 epu = length on micrometer ÷ number of eyepiece divisions.
5. Repeat for **every objective** (each has its own value).

Observe and measure cheek cells

6. Gently scrape the inside of your cheek with a clean cotton bud and smear it on a slide.

7. Add a drop or two of **methylene blue** (stains the nuclei), lower a coverslip at an angle with a mounted needle to avoid air bubbles, and blot off excess stain.
8. Focus with the **low-power** objective first (coarse focus, moving the lens away from the slide), then switch to high power and use the **fine focus** only.
9. Measure several cells in eyepiece units and convert with the calibration value. Calculate a mean.
10. Draw a few cells, label them, and add a title, the magnification and a scale bar.

WORKED EXAMPLE

calibrating and measuring

At ×40 objective, **10 micrometer divisions (100 µm)** line up with **40 eyepiece divisions**.

1 epu = 100 µm ÷ 40 = **2.5 µm**.

A cheek cell measures 24 epu across → 24 × 2.5 = **60 µm**.

At the ×10 objective each eyepiece division covers 4 times more: 1 epu = **10 µm**. That's why you recalibrate when you change objective.

WORKED EXAMPLE

drawing a scale bar

Your drawing is at ×800. You want a scale bar for **10 µm**.

Length of bar = 10 µm × 800 = 8000 µm = **8 mm**. Draw an 8 mm line and label it "10 µm".

Drawing rules

- Sharp HB pencil; **clear, continuous lines**; **no shading** or colouring.
- Draw **only what you see**, large, with proportions correct.
- **Ruled** label lines, no arrowheads, each **touching** the structure, not crossing each other.
- Title, magnification and/or a scale bar. A cell drawing shows individual cells; a **plan diagram** shows only tissue outlines (CP7 in Topic 4).

Safety, errors and improvements

- Cheek swabs: each used once by one person, then straight into **disinfectant**. Methylene blue stains skin and is an irritant: gloves and eye protection.
- Never point the mirror at direct sunlight.
- Errors: forgetting to recalibrate at a new objective; cells overlapping or folded; judging cell edges. Improve by measuring **many cells** and using a **mean**.

Calibrating a graticule

1. Line up the eyepiece graticule with the stage micrometer at the objective you will use.
2. Find a point where the scales match: for example, 40 eyepiece divisions cover 10 stage divisions, which is 10 × 10 µm = 100 µm.
3. One eyepiece division = 100 µm ÷ 40 = 2.5 µm at this magnification.
4. Remove the micrometer, put the specimen on the stage and measure it. A cell 24 divisions long is 24 × 2.5 = 60 µm long.

If you change objective, the calibration must be repeated, because each division now covers a different real length.

EXAM TIP

CP5 has not been examined as a whole practical in WBI12 (2019–2026). Its skills come up constantly through magnification and scale-bar calculations (3.7). Learn the calibration as a four-line method.

COMMON MISTAKE

Quoting a graticule reading in µm without calibrating. Calling the stage micrometer "the graticule". Shading drawings or using arrows on label lines.

Quick checks

QUICK CHECK

Which organelle adds carbohydrate to proteins to form glycoproteins before they are secreted?

- Rough endoplasmic reticulum
- Golgi apparatus
- Lysosome
- Nucleolus

The Golgi apparatus modifies proteins, including adding carbohydrate, and packages them into secretory vesicles. The RER is the commonest wrong answer: it receives and folds the polypeptide, but the main modification and packaging happen in the Golgi.

QUICK CHECK

Which structure is found in a prokaryotic cell but never in a eukaryotic animal cell?

- Ribosome
- Cell surface membrane
- Cytoplasm
- Peptidoglycan cell wall

Animal cells have no cell wall, and only bacteria have walls made of peptidoglycan. Ribosomes are present in both cell types; they differ in size (70S and 80S) but are not absent from eukaryotes.

QUICK CHECK

Why can an electron microscope show ribosomes when a light microscope cannot?

- Electrons have a much shorter wavelength than light, so the resolution is far higher
- An electron microscope magnifies more, and magnification alone reveals detail
- Ribosomes only exist in dead, fixed cells
- Stains used in light microscopy destroy ribosomes

Detail depends on resolution, which depends on wavelength. The shorter wavelength of electrons lets points less than 1 nm apart be distinguished. Extra magnification without better resolution only enlarges a blurred image, which is why the second option is wrong.

QUICK CHECK

A cell is 40 µm in diameter. In a drawing it is 80 mm across. What is the magnification of the drawing?

- ×2
- ×200
- ×2000
- ×20 000

80 mm = 80 000 µm, and 80 000 ÷ 40 = 2000. Forgetting to convert millimetres to micrometres gives ×2, the commonest error.

QUICK CHECK

Forty eyepiece divisions line up with 0.2 mm on a stage micrometer. How long is a cell that spans 15 eyepiece divisions?

- 5 µm
- 50 µm
- 75 µm
- 200 µm

0.2 mm = 200 µm, so one eyepiece division = 200 ÷ 40 = 5 µm. The cell is 15 × 5 = 75 µm. The answer 5 µm is the value of a single division, not the length of the cell.

QUICK CHECK

Which statement correctly distinguishes an organ from a tissue?

- An organ is made of one type of cell; a tissue contains several
- A tissue is found only in animals; organs occur in plants and animals
- An organ contains several different tissues working together; a tissue is a group of similar cells with one function
- Tissues carry out several functions; an organ carries out only one

A tissue is a group of similar cells doing a particular job, while an organ is built from several tissues. The first option reverses the two definitions, and plants have tissues too, such as xylem and phloem.